

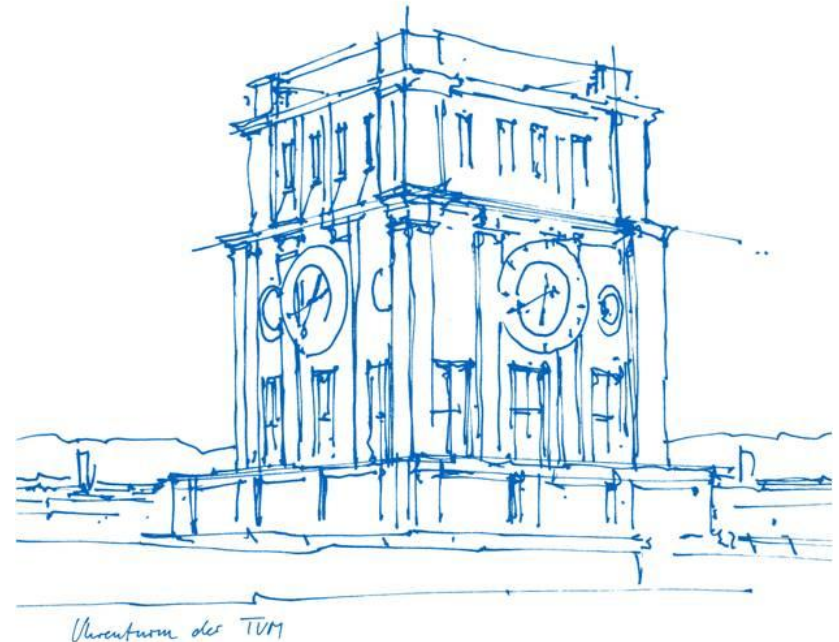
Enterprise Architecture Management and Reference Models: **ArchiMate 3.0 Structure Elements**

Prof. Luise Pufahl

Technische Universität München

TUM School CIT

Information Systems



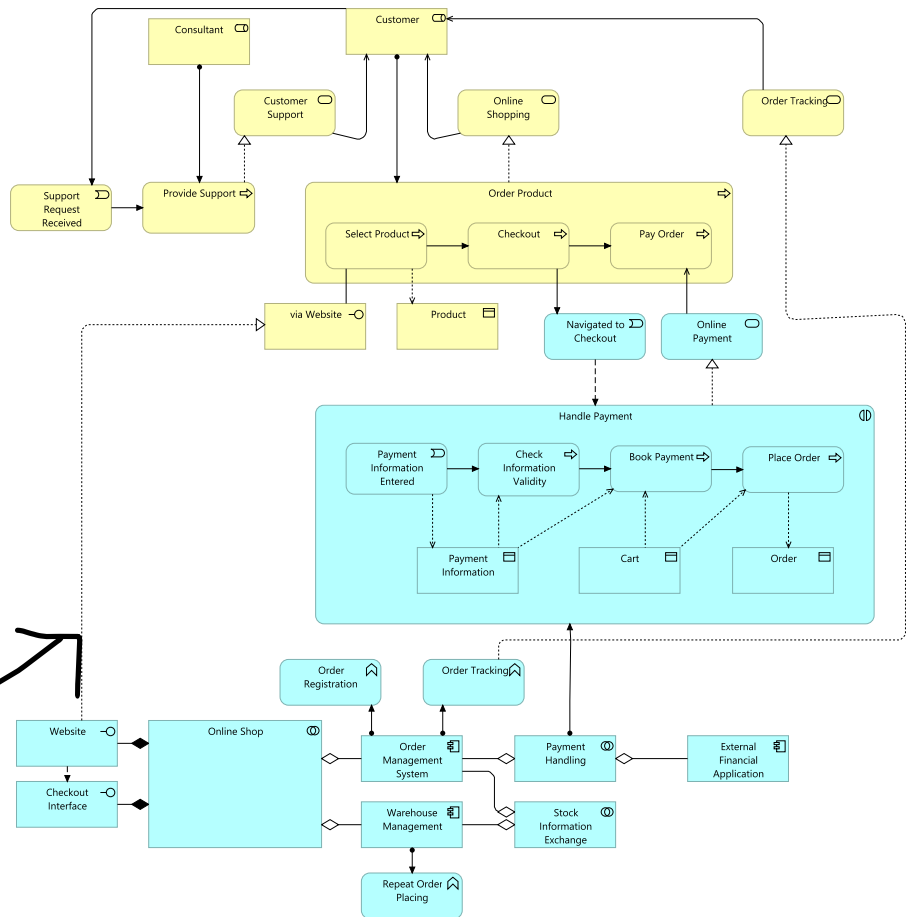
Recap

- What is a system and its architecture?
- What is the difference between an enterprise architecture and an enterprise architecture framework?

What will you learn today?

- Enterprise architecture modeling using the ArchiMate Framework
- ArchiMate Modeling Notation and Association
- Structure Elements

After the next lecture, you will be able to read this enterprise architecture model



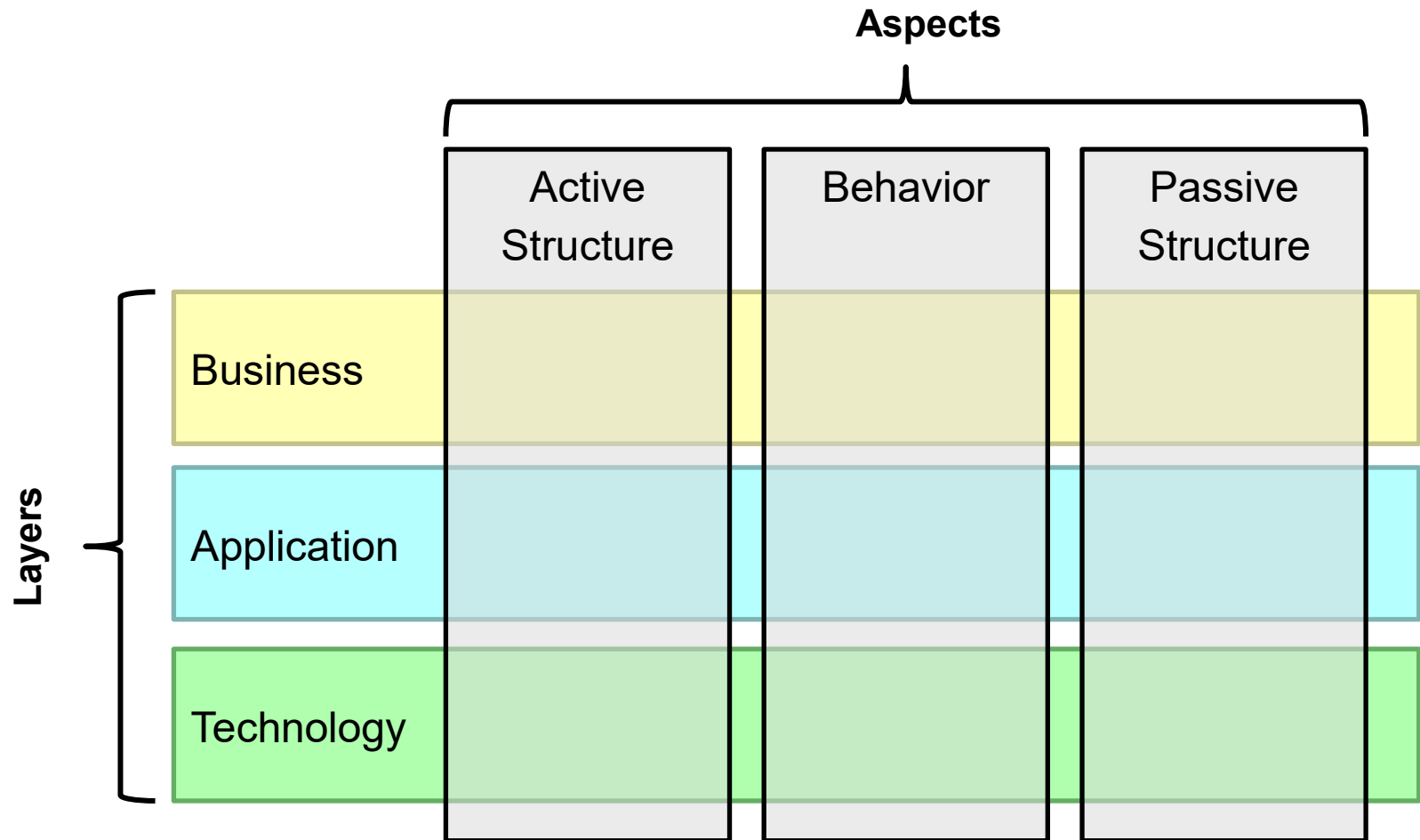
What is ArchiMate?

- Enterprise (architecture) modeling
 - Supports description, analysis, and visualization of enterprise architecture
 - Focus on large, individual systems
- ArchiMate is a **modeling language** for enterprise (architecture) modeling
 - It provides a **graphical notation** of concepts to describe architectures and a **framework** to organize these concepts
 - Syntax based on **colors**, **icons**, and **relationships**
 - It is an open standard maintained by The Open Group:

https://university.sk/wp-content/uploads/2020/01/ARCHIMATE_v3_1_specifikacia.pdf

- Notation inspired by UML 2.0 standard
- Influences from other modeling languages, such as BPMN

ArchiMate Framework (I/II)

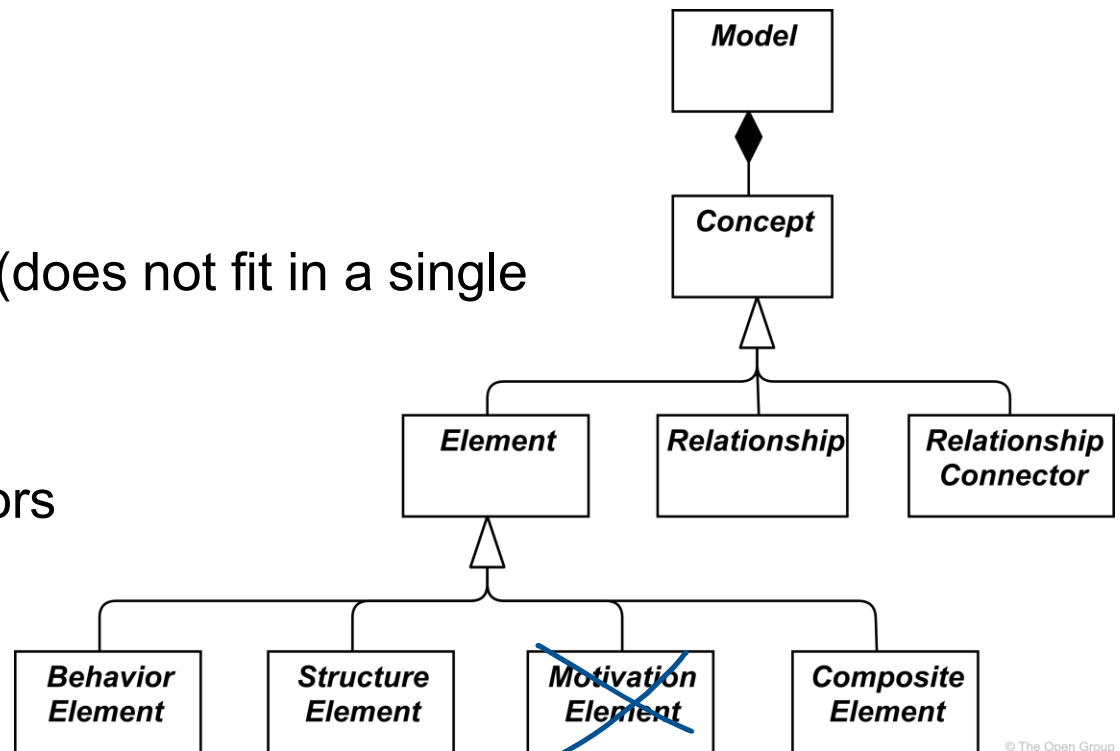


ArchiMate Framework (II/II)

- Aspects
 - Active structure elements are subjects that can perform behavior
 - Behavior elements represent dynamic aspects of the enterprise
 - Passive structure elements can be accessed by behavior elements
- Layers
 - Business Layer models the operational organization of an enterprise in a technology-independent manner
 - Application Layer models the application architecture (structure, behavior, and interaction of applications used in the enterprise)
 - Technology Layer models the technology infrastructure of the enterprise (structure and behavior, hard- and software)

ArchiMate Hierarchy of Concepts

- ArchiMate Models are a collection of concepts
- Concepts
 - Elements
 - Behavior Element
 - Structural Element
 - Composite Element (does not fit in a single aspect)
 - Relationships
 - Relationship Connectors



© The Open Group

not relevant in this lecture

ArchiMate Structure Elements

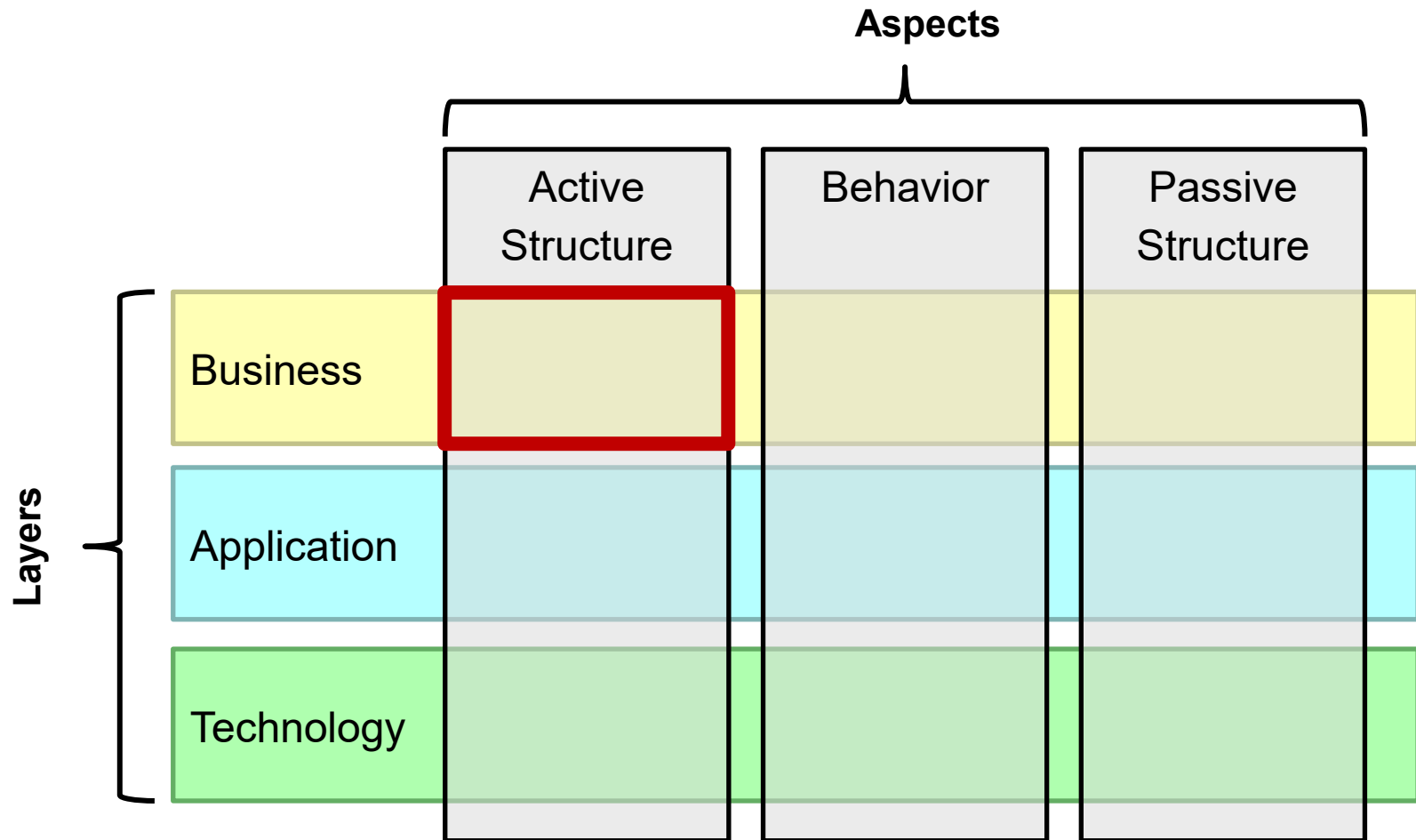
Structure Elements

- Structure elements represent *subjects that can perform behavior*
 - Actors and Roles
 - Collaborations
 - Application Components and Devices
 - ...
- Active Structure Elements
 - Internal Active Structure Elements represent an entity that is capable of performing a behavior
 - External Active Structure Elements, called Interfaces, represent a point of access where one or more services are provided to the environment
- Passive Structure Elements
 - Can be accessed by behavior elements
 - Elements on which behavior is performed

ArchiMate Structure Elements

Business Layer

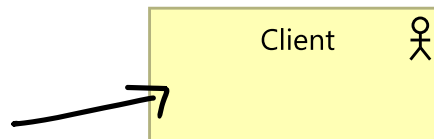
ArchiMate Framework



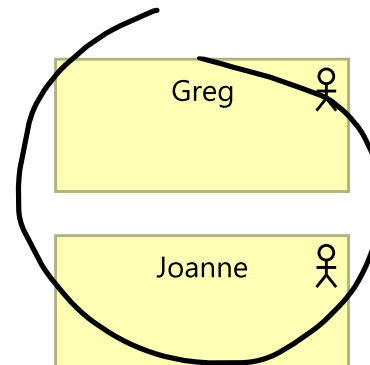
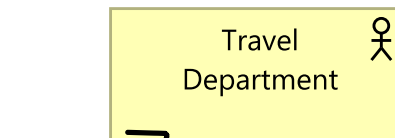
Active Structure Elements in Business Layer

- **Business Actor** can perform behavior
 - Persons (customer or employee)
 - Group of people or resources (department, business unit)
 - Has long-term status in organization
 - Internal active structure element
 - Labeling: Noun
- Examples:

A client performs behavior in an insurance company, e.g., claim handling



Department that checks travel requests



Specific persons can also be an actor

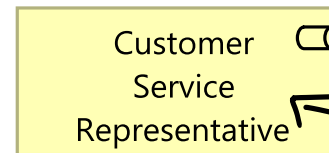
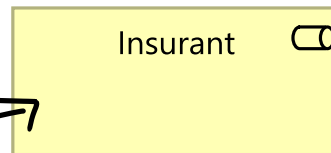
Reminder: Business Layer = operational view + technology-independent

Active Structure Elements in Business Layer

- **Business Role** has responsibility to perform behavior
 - The part an actor plays in certain actions or events
 - multiple actors can fulfill the same role and one actor can fulfill multiple roles
 - Roles are more generic than actors
 - Internal active structure element
 - Labeling: Noun

- **Examples:**

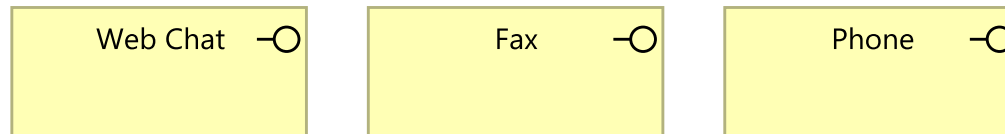
Insurant is more generic, a client may have multiple roles



Many actors can have this role

Active Structure Elements in Business Layer

- **Business Interface** represents a point of access where business services are made available to the environment (provides behavior)
 - External active structure element
 - Same service can be offered through various interfaces (e.g., by email, phone, ...)
 - Labeling: Noun
- Examples: A contact center has several points of access where customers can get in touch with the company



Active Structure Elements in Business Layer

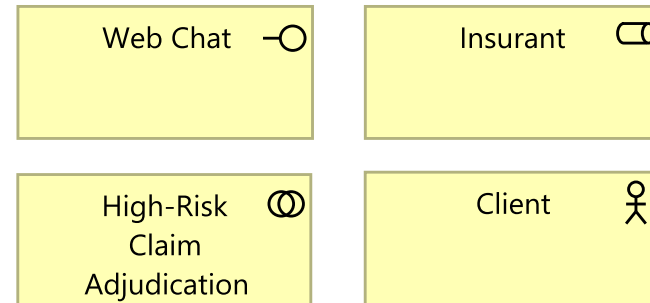
- **Business Collaboration** represents an aggregate of two or more business internal active elements that work together
 - Internal active structure element
 - Performed behavior is collective behavior
 - Labeling: Noun
- Examples: In an insurance company, high-risk claim adjudications have to be handled by a specialist and a customer service representative (= collaboration)

High-Risk
Claim
Adjudication



Active Structure Elements in Business Layer

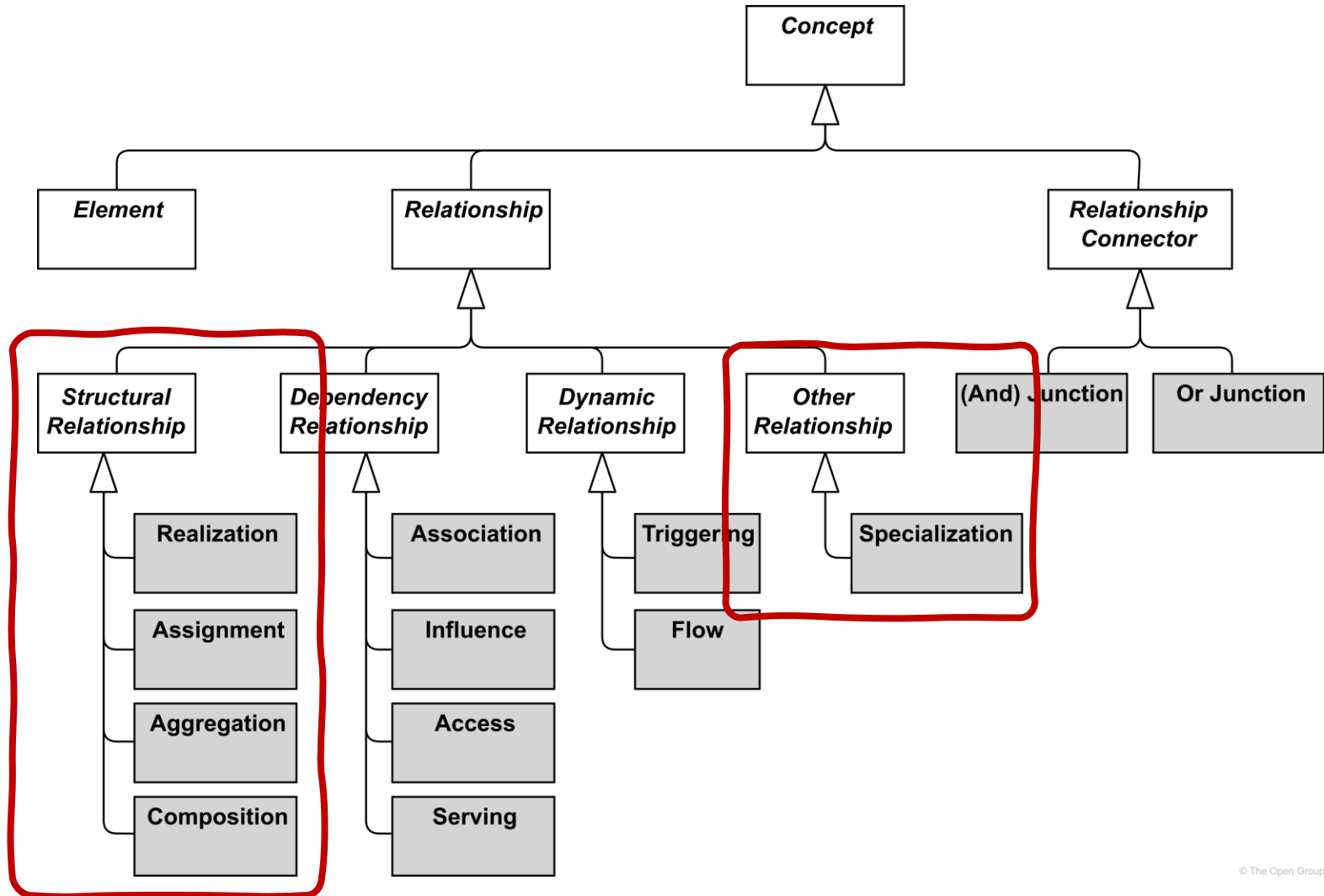
- Overview
 - Business Actor
 - Business Role
 - Business Collaboration
 - Business Interface



- How do these active structure elements relate to each other?

ArchiMate Relationships (I/II)

Overview of Relationships



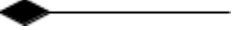
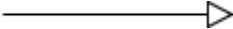
© The Open Group

ArchiMate

Structural Relationships

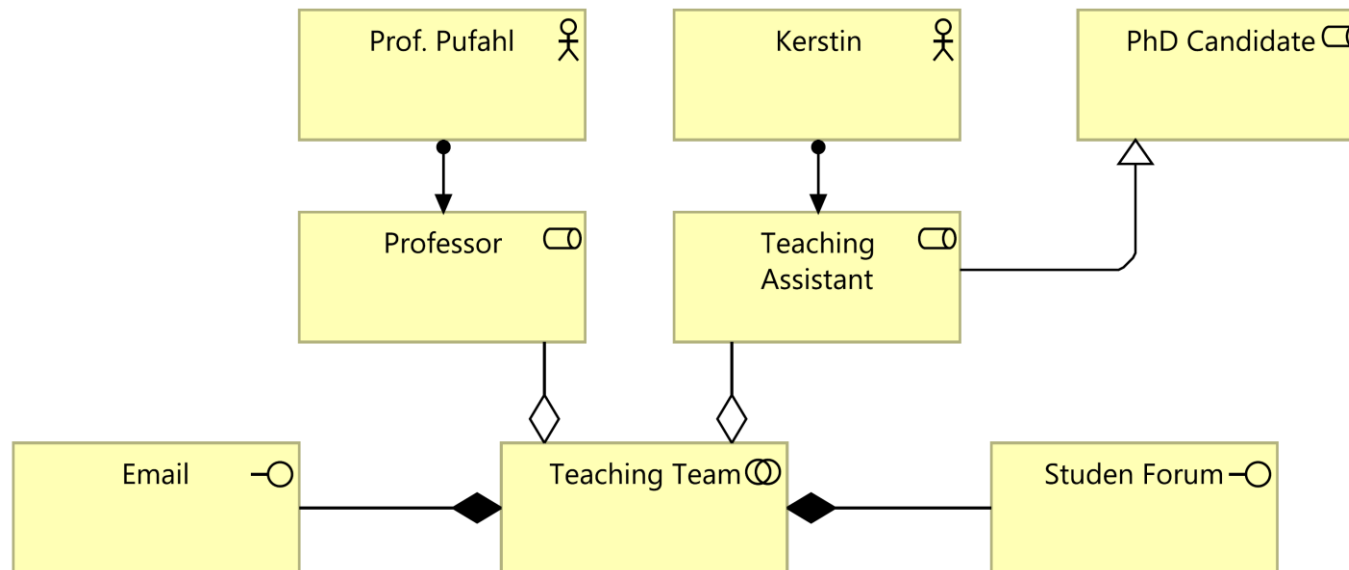
Structural Relationships and Specialization

- Model static construction or composition of the same or different types of elements

Relationship	Description	Notation
Realization	Represents that an element plays a critical role in the creation, achievement, sustenance, or operation of a more abstract element	
Assignment	Represents the allocation of responsibility, performance of behavior, storage, or execution	
Aggregation	Represents that an element groups one or more elements	
Composition	Represents that an element consists of one or more other elements	
Specialization	Represents that an element is a particular kind of another element	

Example: Structural Relationships

- A lecturer team consists of a professor and a teaching assistant
- They can be contacted via email or via the student forum



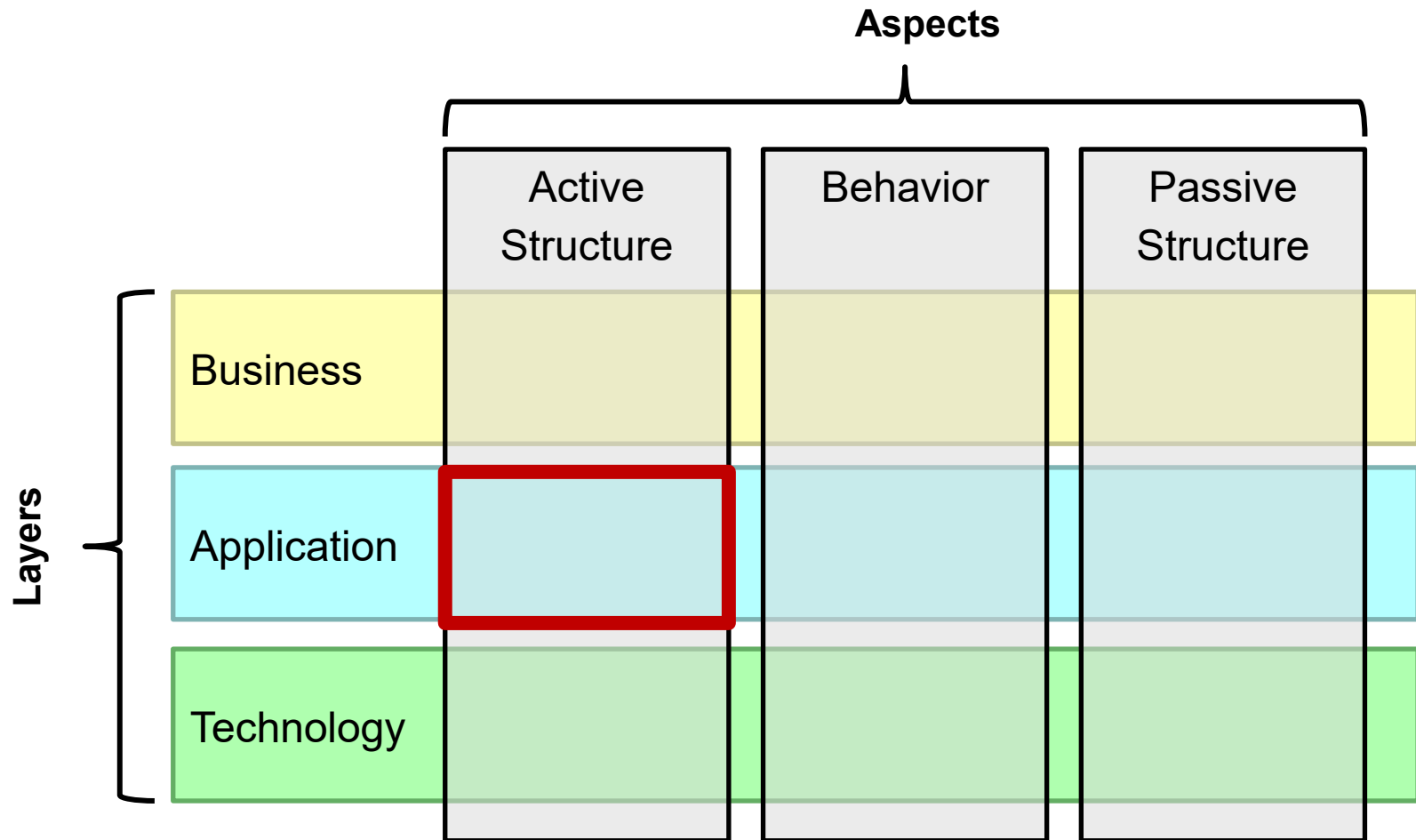
Lecture Quiz L05Q01

You find the quiz on Moodle

ArchiMate Structure Elements

Application Layer

ArchiMate Framework (I/II)

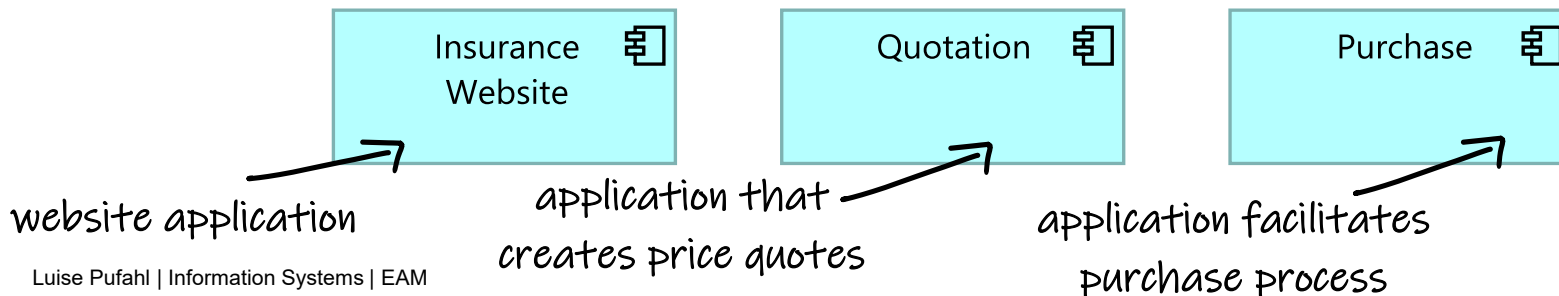


Active Structure Elements in Application Layer

- **Application Component** represents an encapsulation of application functionality aligned to implementation structure
 - Modular and replaceable
 - Software that will perform the necessary functions
 - Labeling: Noun

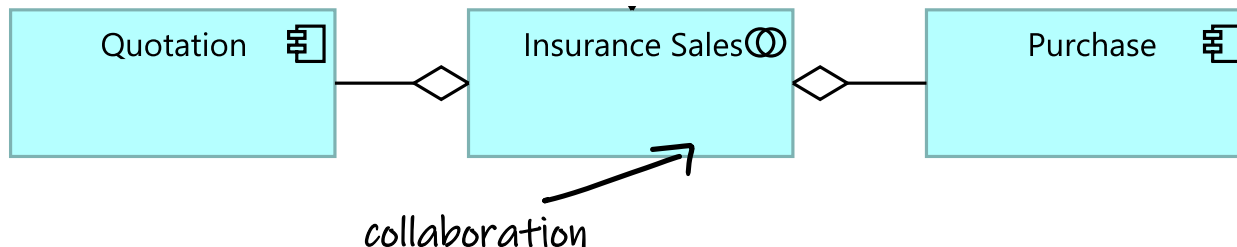
Reminder: Application Layer = structure, behavior, interaction of applications

- Examples: software applications, sub-applications, information systems



Active Structure Elements in Application Layer

- **Application Collaboration** aggregates at least two application components to fulfill a certain, collective behavior
 - Similar to business collaboration element
 - Labeling: Noun
- Examples: The insurance sales application is the collective behavior of the quotation application (generating quotes) and purchase application (procurement of customer after quote is accepted)



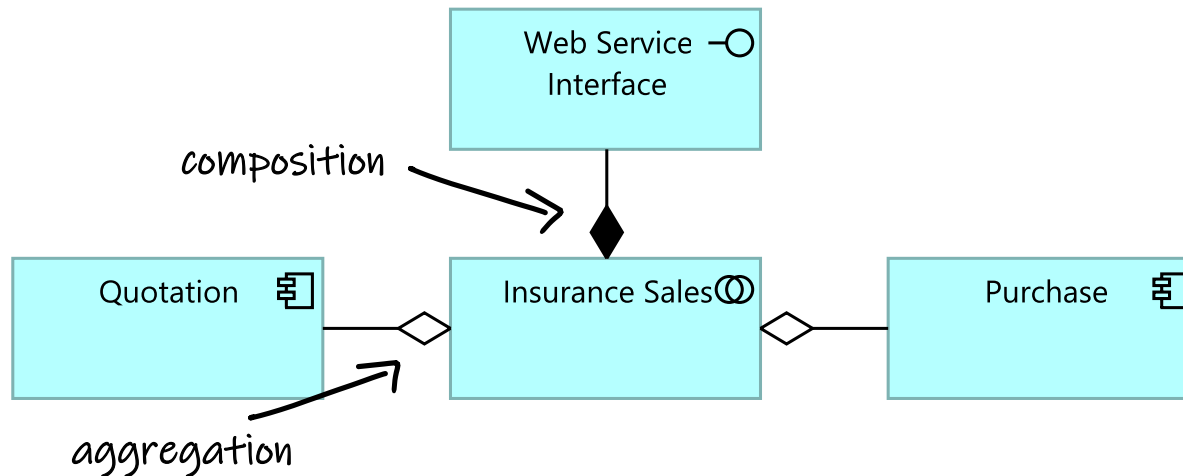
Active Structure Elements in Application Layer

- **Application Interface** represents a point of access where an application service is made available to a user or an application component
 - Specifies how the functionality of an application component can be accessed by other elements
 - Can be part of an application component (composition) if the interface is provided by that component and serves others
 - Can be assigned to application services if the interface exposes these services to the environment
 - Labeling: Noun
- Examples: Web service interfaces allow different applications to interact and exchange data



Active Structure Elements in Application Layer

- Overview
 - Application Component
 - Application Collaboration
 - Application Interface
- What about relations between those elements?

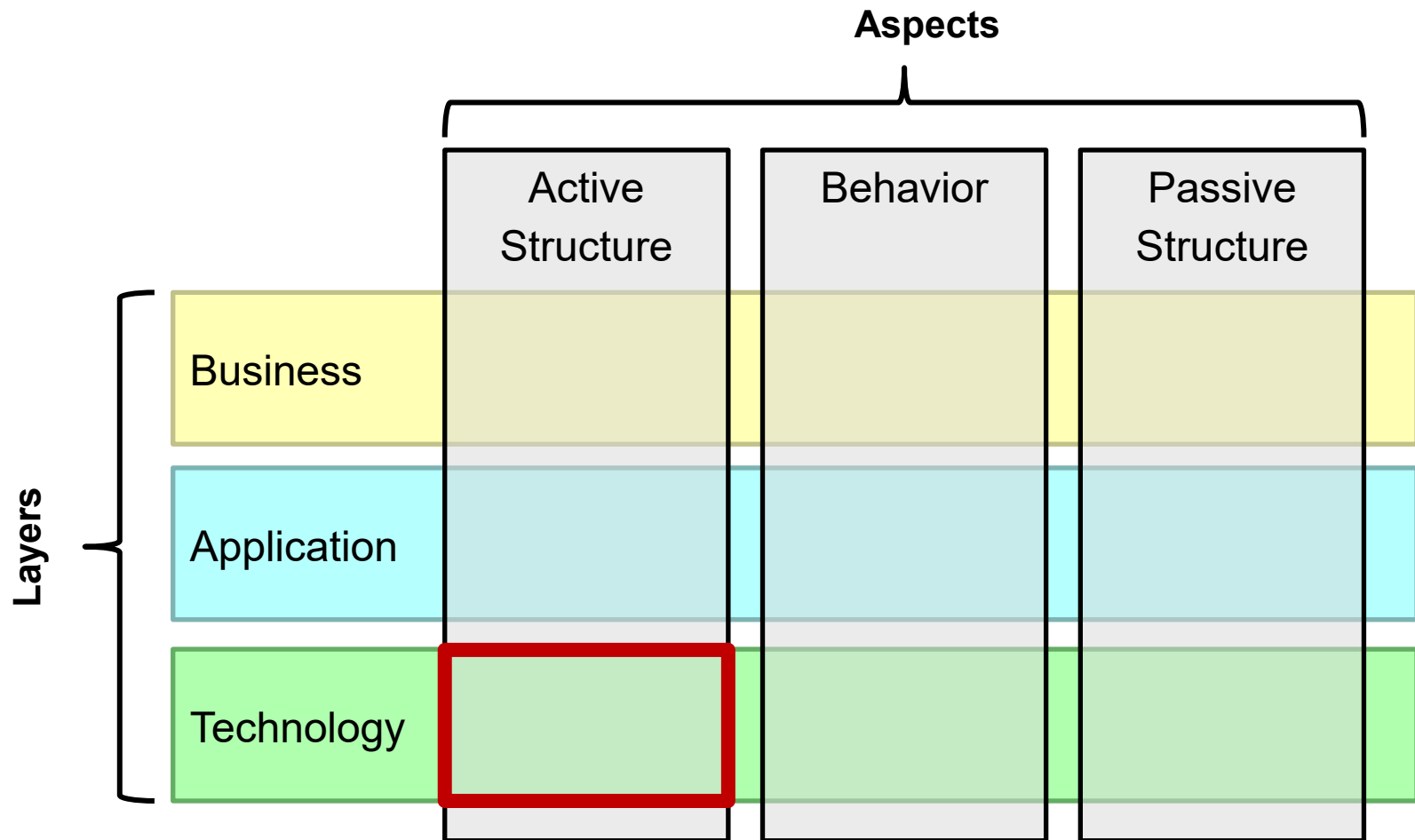


Lecture Quiz L05Q02

You find the quiz on Moodle

ArchiMate Structure Elements Technology Layer

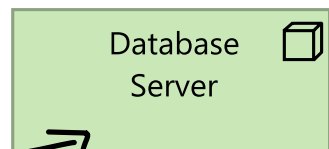
ArchiMate Framework (I/II)



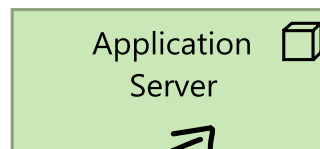
Active Structure Elements in Technology Layer

- **Node** represents a computational or physical resource that hosts, manipulates, or interacts with other computational or physical resources
 - Perform technology behavior
 - Used to model application platforms
 - Labeling: Noun
- Examples: Typical nodes of an enterprise are

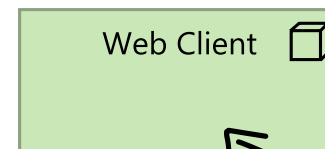
Reminder: Technology Layer = low level hardware, software and connectivity



hosts the database and enables access to data



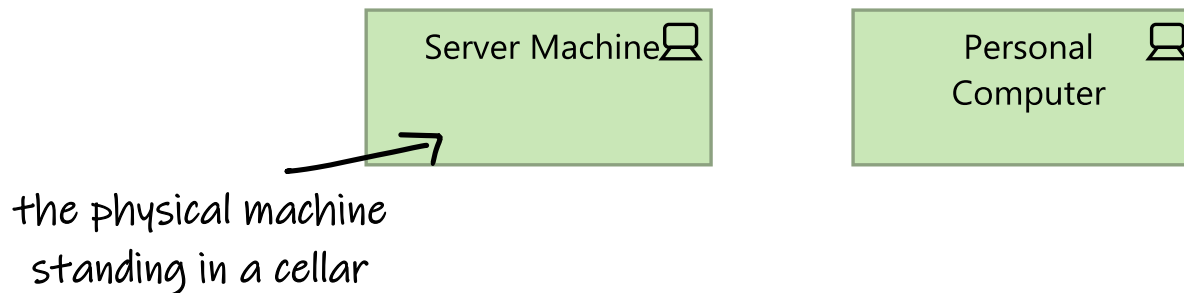
hosts different applications of the enterprise, interacts with the database



allows users to navigate the internet, interacts with application server

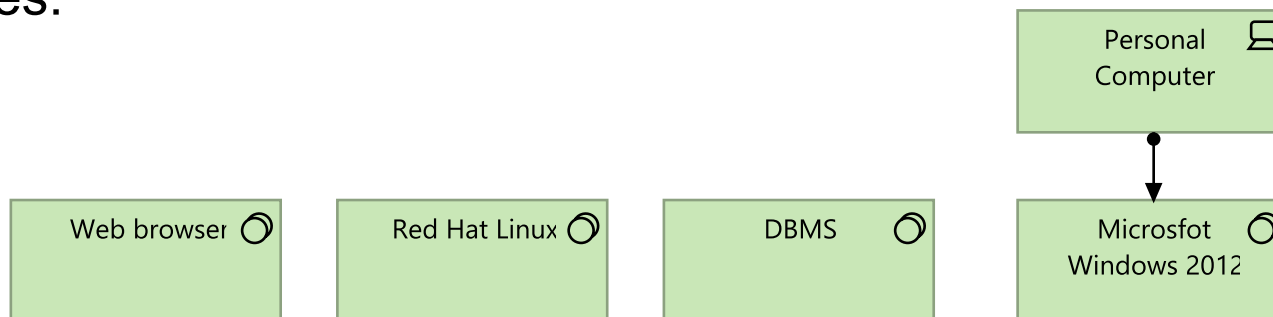
Active Structure Elements in Technology Layer

- **Device** represents a physical IT resource upon which system software and artifacts may be stored or deployed for execution
 - Has processing capability
 - Used to model hardware systems, such as PCs, smartphones, etc.
 - Labeling: Noun referring to the type of hardware
- Examples:



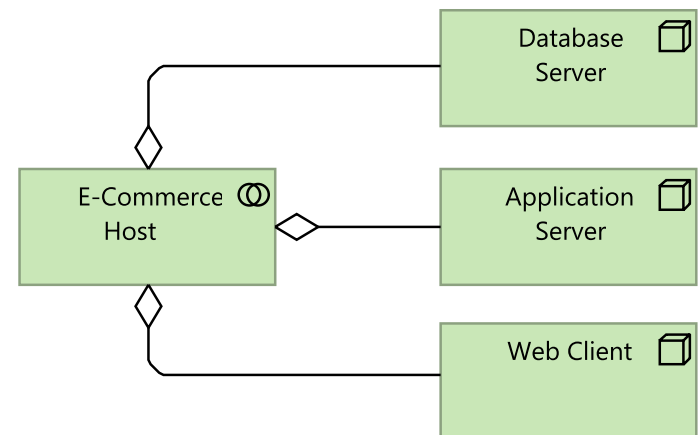
Active Structure Elements in Technology Layer

- **System Software** represents software that provides or contributes to an environment for storing, executing, and using software or data deployed within it.
 - Low level software: system software runs artifacts
 - Usually combined with a device
 - Operating systems, workflow engines, DBMS
 - Labeling: Noun referring to the type of execution environment
- Examples:



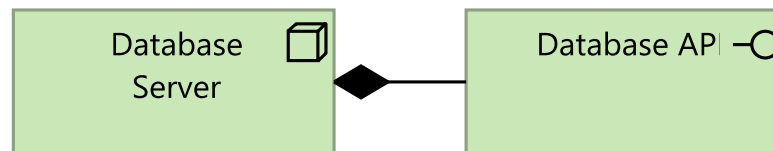
Active Structure Elements in Technology Layer

- **Technology Collaboration** aggregates two or more technology internal active structure elements that work together to perform collective technology behavior
 - Models logical or temporary collaboration
 - Does not exist as a separate entity
 - Collaborations can be used to represent clustered infrastructures
 - Labeling: Noun
- Examples: For E-Commerce Hosting,
 - the database server manages orders and account profiles
 - the application server generates dynamic content
 - the web client provides the website



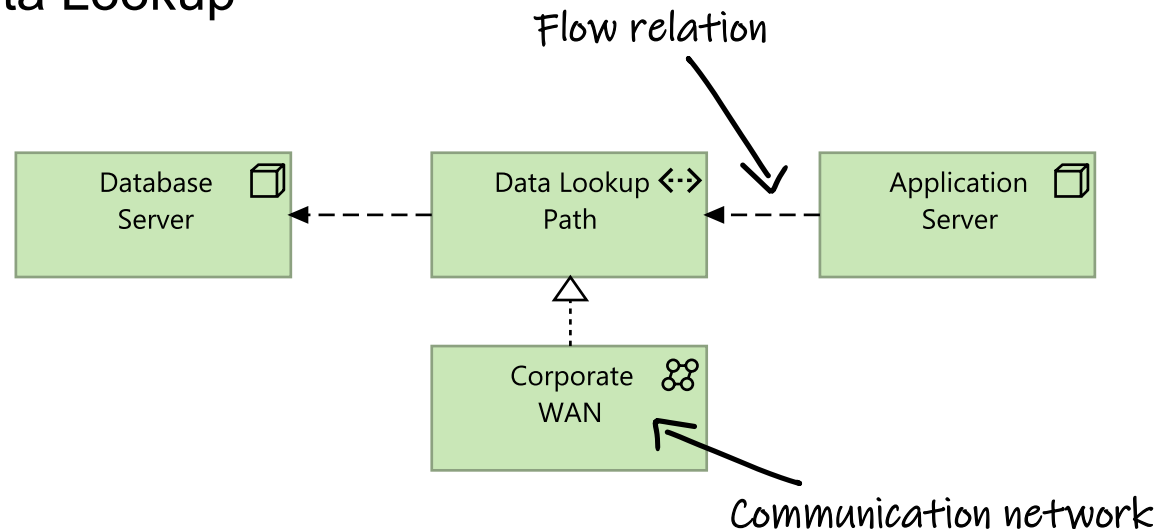
Active Structure Elements in Technology Layer

- **Technology Interface** represents a point of access where technology services offered by a technology internal active structure element can be accessed
 - Can be part of an active structure element (composition) if the interface is provided by that element and serves others
 - Can be assigned to technology services if the interface exposes these services to the environment
 - Labeling: Noun
- Example: APIs



Active Structure Elements in Technology Layer

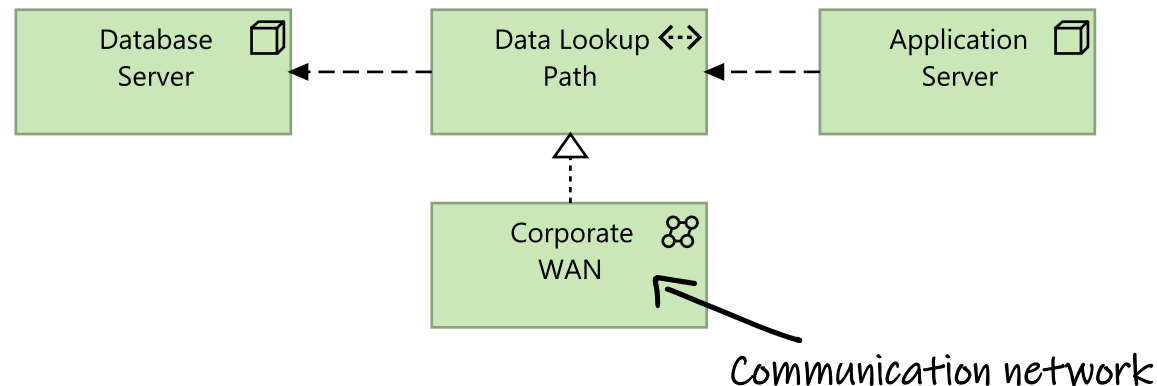
- **Path** represents a link between two or more nodes, through which these elements can exchange data.
 - Models logical communication relations
 - Realized by one or more communication networks
 - Labeling: Noun
- Example: Data Lookup



Active Structure Elements in Technology Layer

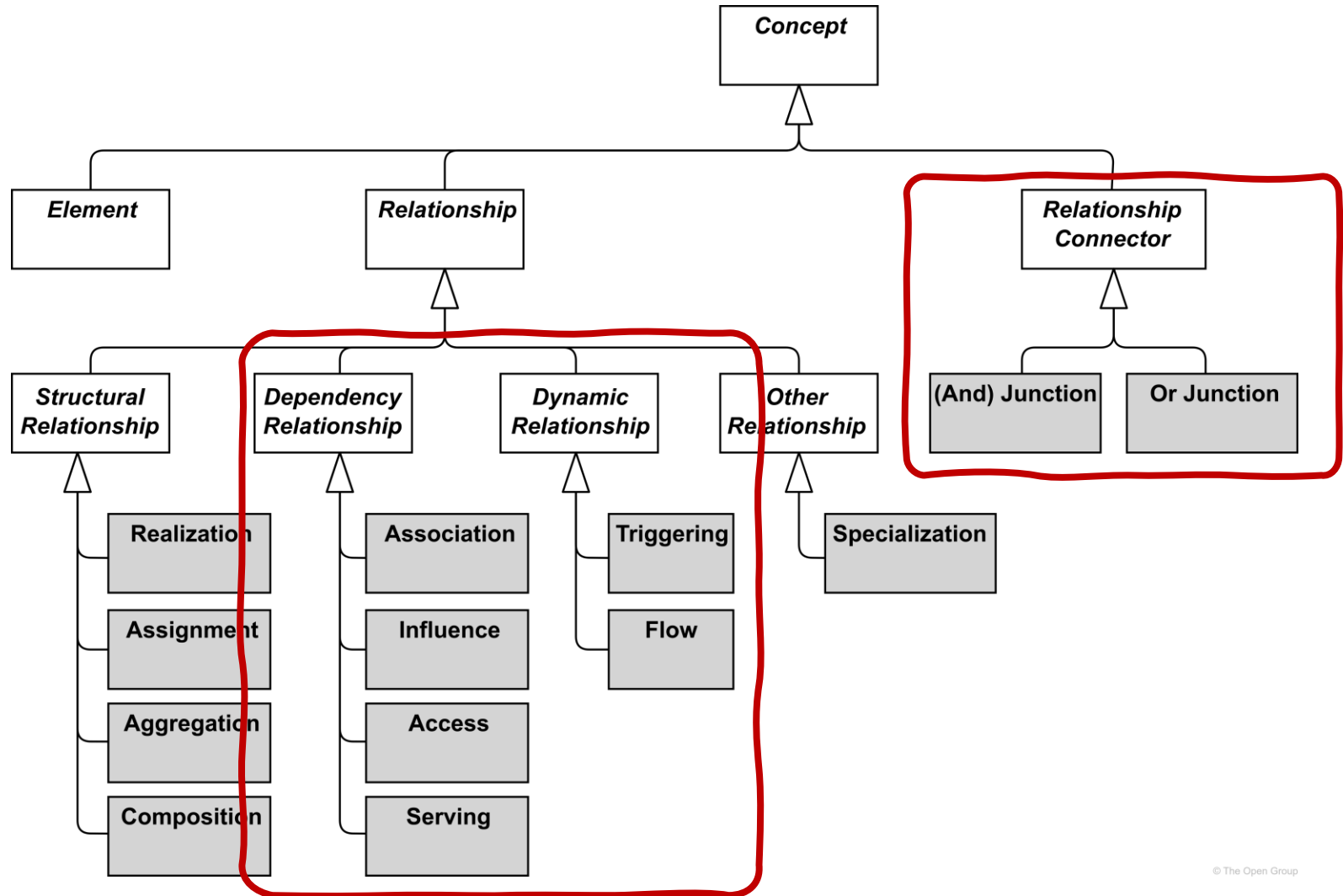
- **Communication network** represents a set of structures that connects devices or system software for transmission, routing, and data reception
 - Most basic network: single link between devices
 - Physical realization of the logical paths between nodes
 - Can aggregate devices and system software to model the routers, switches, firewalls, etc. that are part of the network
 - Labeling: Noun

- Example:



ArchiMate Relationships (II/II)

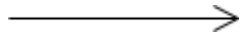
Overview of Relationships



© The Open Group

Dependency Relationships

- Describe how elements support or are used by other elements

Relationship	Description	Notation
Serving	Represents that an element provides its functionality to another element	
Access	Represents the ability of behavior and active structure elements to read and write passive structure elements	
Influence	Represents that an element affects the implementation of a motivation element Not relevant for this lecture	
Association	Represents an unspecified relationship	

Dynamic Relationships

- Describe temporal dependencies between elements within the architecture

Relationship	Description	Notation
Triggering	Represents a temporal or causal relationship between elements	
Flow	Represents transfer from one element to another	

Relationship Connectors (Junctions)

- A **junction** connects relationships of the same type
 - It may have multiple incoming and one outgoing relationship or multiple one incoming and multiple outgoing relationships
 - A junction with multiple incoming and outgoing relationships is considered a shorthand of two contiguous junctions
- We differentiate between:

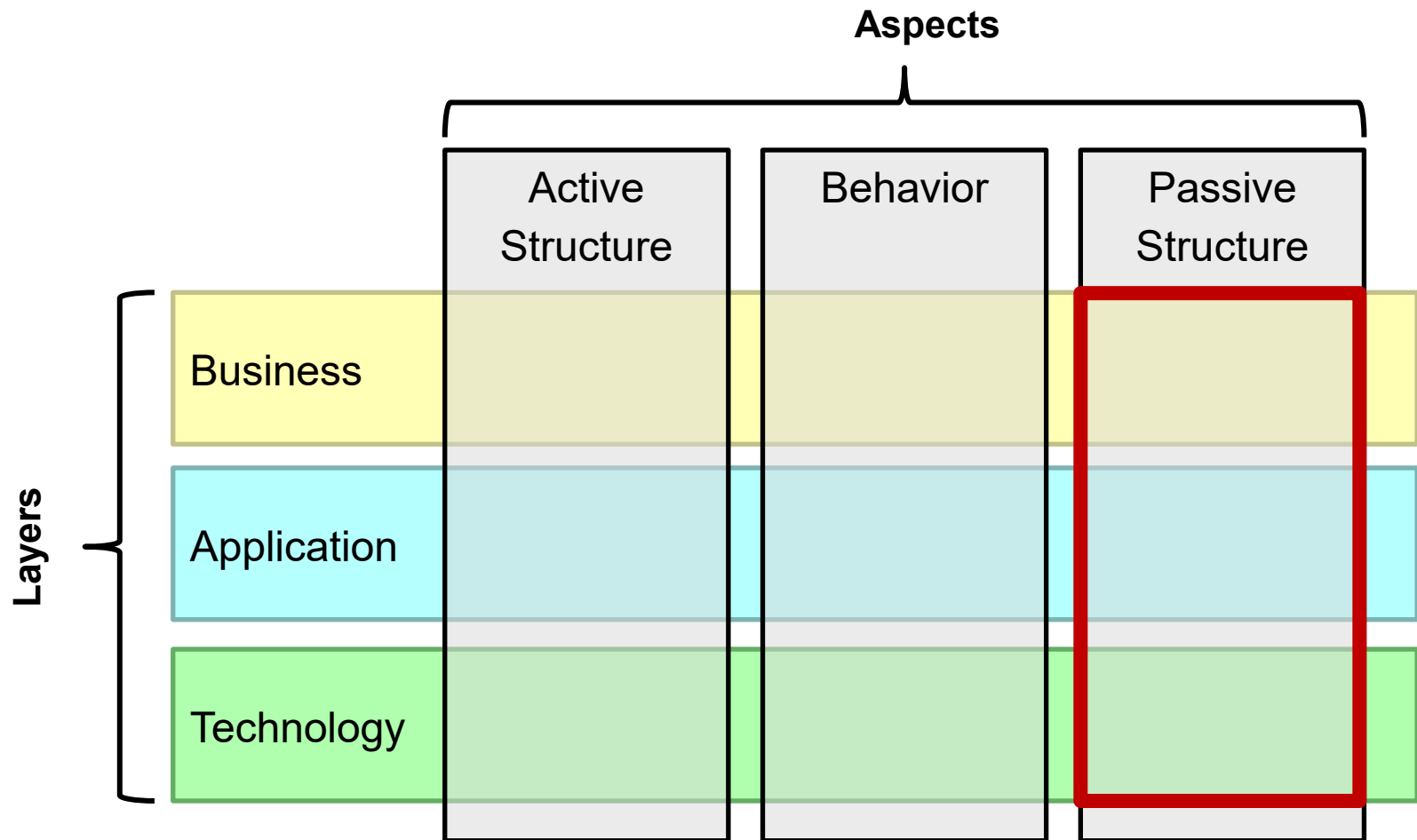
OR Junction ○	AND Junction ●
– At least one element participates in the relationship – Expresses both inclusive and exclusive conditions (can be indicated by naming the junction)	– All elements together must participate in the relationship

Lecture Quiz L05Q03

You find the quiz on Moodle

ArchiMate Passive Structure Elements

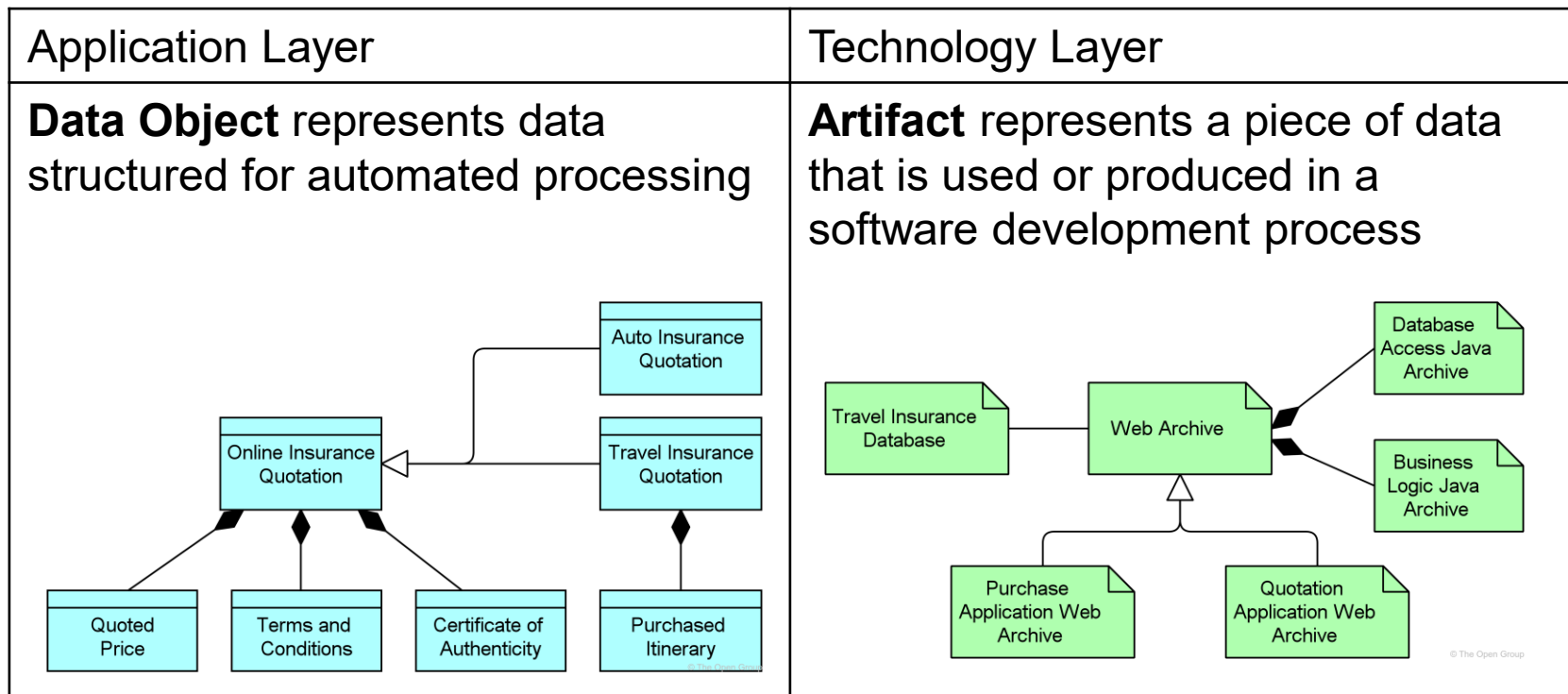
ArchiMate Framework (I/II)



Passive Structure Elements

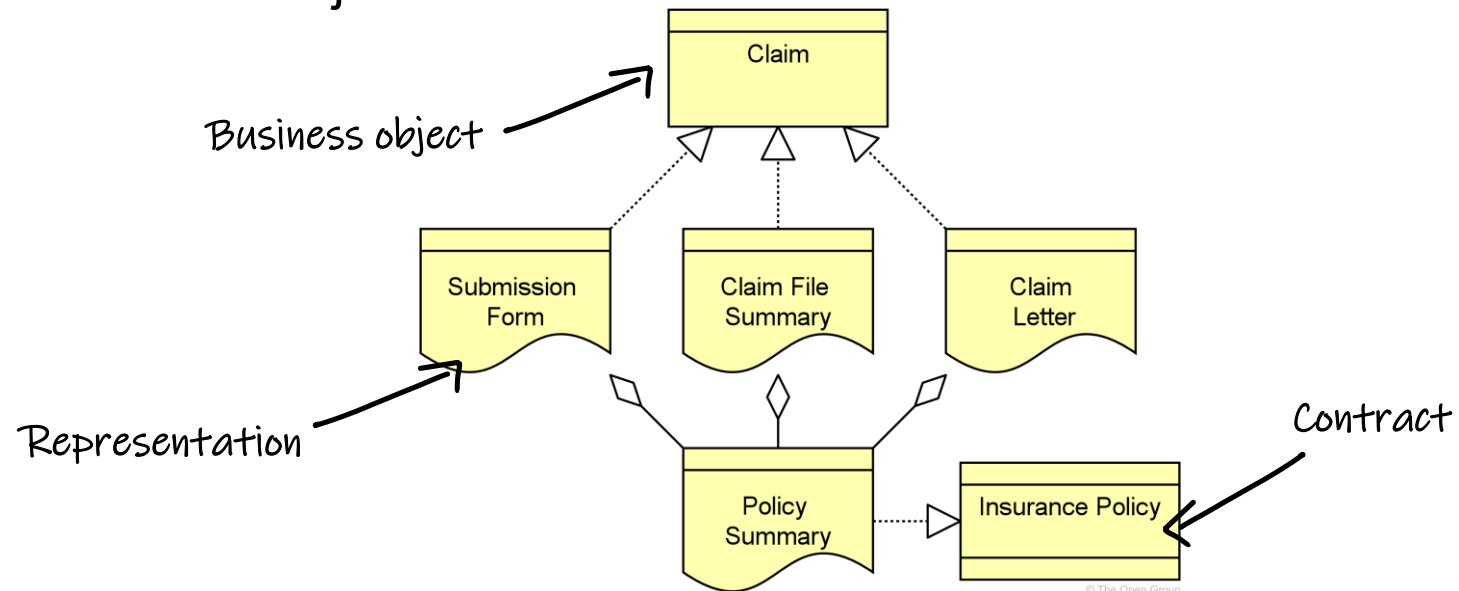
Next lecture

- Passive structure elements are manipulated by behavior elements
- Application and Technology Layer only have one passive structure element



Passive Structure Elements in Business Layer

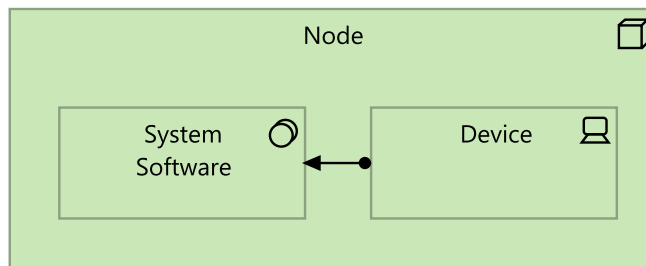
- **Business Object** represents a concept used within a particular business domain
- **Contract** represents a specification of an agreement between a provider and a consumer
- **Representation** represents a perceptible form of the information carried by a business object



Nesting

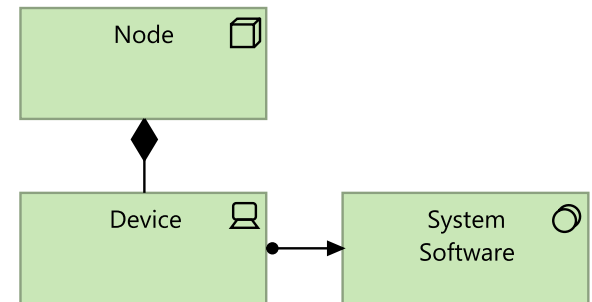
Nesting of Elements

- **Structural relationships** may also be expressed by nesting the united concept with the uniting element
- Regarding dependency relationships, the **access relationship** can be nested
- **Specialization** can also be expressed by nesting the specialized element inside the generic element
- Might lead to ambiguous views



Nested structure

=

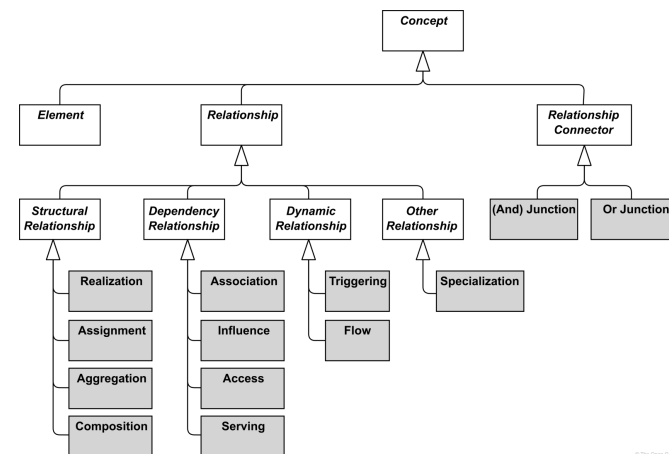
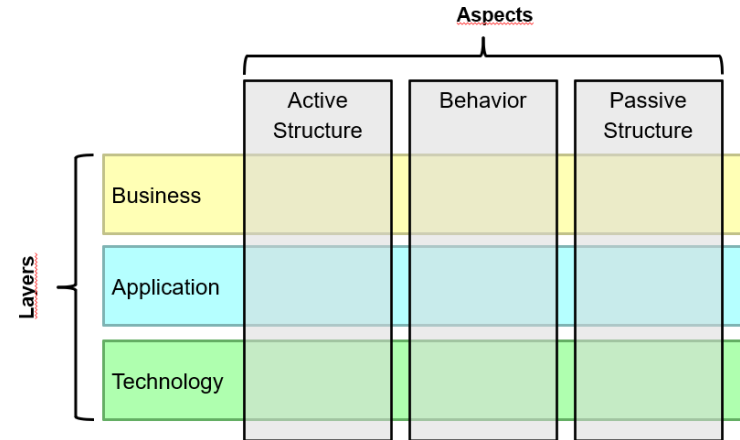


Explicit structure

Summary & Announcements

Summary

- ArchiMate Framework
- Different Layers
 - Business Layer
 - Application Layer
 - Technology Layer
- Notation for structure elements
 - Active structure elements
 - Passive structure elements
- Relationships
- Nesting



© The Open Group

Announcements

- Homework Exercise H04 published on Moodle
- Two tasks
 - ArchiMate Modeling Business and Application Layer

Enterprise Architecture Management and Reference Models
Prof. Luise Pufahl, Kerstin Andree



Homework Exercise H04

Homework exercises support you in practicing the content taught in the lecture. The learning goal is to remember and understand important concepts. Some tasks also target the application of taught methods and concepts. We highly recommend working on the exercises using pen and paper to practice for the exam. Keep your answers concise by using the terminology introduced in the lecture. Try not to check the solution before or while solving the exercises.

H04E01 – Freddy Furious' Bike Shop Business Layer (15 points)

Please consider the following case example of Freddy Furious' bike shop:

Frederic "Freddy" Furious is the proud owner of a bicycle shop that offers various services and products for cycling enthusiasts. Whether you prefer the convenience of online shopping or the hands-on experience of visiting the physical store, Freddy's shop has you covered.

The shop provides a range of bicycles available for purchase, both online and onsite. If you decide to buy a bike in person, knowledgeable shoppers are ready to assist you in making the right

Announcements

- Skill Check tomorrow, 21.05.2026 at 10.00 in Etzelstraße 38.01.1
 - Bring a pen and your student ID
 - Be on time!
 - Seat Plan published via Moodle
- Next week:
 - L06 ArchiMate II – Behavior Elements
 - Tutorials: No tutorial next week
- Kick-Off Study Project tomorrow, 21.05.2026 (after skill check)
 - Introduction of cases
 - Information on registration, examination, submission